

V10: sampling resolution within CT and CE

512 predictor evaluations minus 128, separately within each method; temperature 0.5, seeds 1700/1701, 100 requests per seed. Four configurations / eight runs / 800 requests, all independently rescored. This is a resolution contrast, not CE minus CT and not equal compute: 512 uses four times the model evaluations.

No superiority established. V5/V6 and complete V9 outcomes informed this design. The separately audited V8 CT and completed V8b CE checkpoints are reused without retraining. The original V8 controller/campaign remains failed. Final UDLM seeds 0/1/2 remain reserved. Local MDLM quality 85.8% and paper GenMol V1 84.6% are contextual, differently budgeted comparisons.

Configuration	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t050_n128	repaired	100.000 +/- 0.000	100.000 +/- 0.000	45.500 +/- 7.778	0.89176 +/- 0.00644
ct_t050_n128	strict	72.000 +/- 9.899	100.000 +/- 0.000	35.500 +/- 12.021	0.87538 +/- 0.00160
ct_t050_n512	repaired	99.500 +/- 0.707	100.000 +/- 0.000	45.000 +/- 2.828	0.89072 +/- 0.00114
ct_t050_n512	strict	73.500 +/- 2.121	100.000 +/- 0.000	32.500 +/- 3.536	0.87636 +/- 0.00515
ce_t050_n128	repaired	99.000 +/- 1.414	99.490 +/- 0.722	46.500 +/- 0.707	0.89277 +/- 0.00262
ce_t050_n128	strict	72.500 +/- 6.364	100.000 +/- 0.000	36.000 +/- 0.000	0.87814 +/- 0.00067
ce_t050_n512	repaired	98.500 +/- 0.707	100.000 +/- 0.000	48.000 +/- 4.243	0.89265 +/- 0.00113
ce_t050_n512	strict	71.500 +/- 0.707	100.000 +/- 0.000	38.500 +/- 3.536	0.87789 +/- 0.00508

Per-configuration means +/- sample SD weight seeds equally; no pooling across configurations. Validity and quality divide by requests; uniqueness divides by valid molecules. Quality counts unique valid molecules with QED >= 0.6 and SA <= 4. Diversity is the released PyTDC metric on unique valid molecules. Repaired decoding uses SAFE fix=True and largest-component selection by SMILES length; strict uses fix=False plus RDKit validation.

Paired molecular changes: 512 minus 128

Method / seed	Branch	Delta validity pp	Delta uniqueness pp	Delta quality pp	Delta diversity
CT / 1700	repaired	0.000	0.000	3.000	-0.00640
CT / 1701	repaired	-1.000	0.000	-4.000	0.00433
CT mean +/- SD	repaired	-0.500 +/- 0.707	0.000 +/- 0.000	-0.500 +/- 4.950	-0.00104 +/- 0.00759
CT / 1700	strict	10.000	0.000	3.000	-0.00153
CT / 1701	strict	-7.000	0.000	-9.000	0.00349
CT mean +/- SD	strict	1.500 +/- 12.021	0.000 +/- 0.000	-3.000 +/- 8.485	0.00098 +/- 0.00355
CE / 1700	repaired	-1.000	0.000	4.000	0.00093
CE / 1701	repaired	0.000	1.020	-1.000	-0.00117
CE mean +/- SD	repaired	-0.500 +/- 0.707	0.510 +/- 0.722	1.500 +/- 3.536	-0.00012 +/- 0.00149
CE / 1700	strict	-5.000	0.000	5.000	-0.00337
CE / 1701	strict	3.000	0.000	0.000	0.00287
CE mean +/- SD	strict	-1.000 +/- 5.657	0.000 +/- 0.000	2.500 +/- 3.536	-0.00025 +/- 0.00441

Pairing is by seed/run labels, not molecule-level trajectories: different step counts change random draws and reverse paths. All declared pairs must define a metric before its mean/SD is shown; undefined is never zero. Both methods and negative changes remain disclosed. Two-seed SD is descriptive, not a confidence interval or significance test.

Generation wall time: 512 / 128

Method / seed	128-step seconds	512-step seconds	Ratio / mean +/- sample SD
CT / 1700	15.143	57.415	3.792
CT / 1701	13.748	52.184	3.796
CT mean +/- SD	--	--	3.794 +/- 0.003
CE / 1700	14.948	63.671	4.259
CE / 1701	14.052	53.255	3.790
CE mean +/- SD	--	--	4.025 +/- 0.332

Generation seconds include model sampling/tokenizer decoding plus released SAFE postprocessing; they exclude checkpoint loading and metric scoring. The mean is the equal-seed mean of paired ratios, not a ratio of means. Sequential GPU scheduling, device mapping and external activity can change wall time; four times NFE need not mean four times wall time.

Raw SAFE ring and parenthesis diagnostics

Configuration / seed	Requests	Odd ring label	Bad parentheses	Both flags	Either flag	Flagged strict-valid
ct_t050_n128 / 1700	100	19	14	3	30	0
ct_t050_n128 / 1701	100	8	12	2	18	0
ct_t050_n512 / 1700	100	7	14	1	20	0
ct_t050_n512 / 1701	100	11	11	1	21	0
ce_t050_n128 / 1700	100	12	8	3	17	0
ce_t050_n128 / 1701	100	12	14	4	22	0
ce_t050_n512 / 1700	100	12	12	2	22	0
ce_t050_n512 / 1701	100	11	11	2	20	0

Odd ring label: remove bracket atoms, recognize single-digit, %NN and %(integer) labels, normalize integer labels and flag odd counts. Even counts allow legitimate label reuse and do not establish chemical validity. Parentheses: remove bracket atoms and %(integer) labels, then flag a negative prefix depth or nonzero final depth. Counts overlap.

Method: mean 512-minus-128 count +/- SD	Odd ring	Parentheses	Both	Either	Flagged strict-valid
CT	-4.500 +/- 10.607	-0.500 +/- 0.707	-1.500 +/- 0.707	-3.500 +/- 9.192	0.000 +/- 0.000
CE	-0.500 +/- 0.707	0.500 +/- 4.950	-1.500 +/- 0.707	1.500 +/- 4.950	0.000 +/- 0.000

Released repair behavior: counts across the two 100-request seeds

Configuration	Requests	Recovered: count / requests (%)	Largest-component selection: count / requests (%)
ct_t050_n128	200	56 / 200 (28.0%)	45 / 200 (22.5%)
ct_t050_n512	200	52 / 200 (26.0%)	40 / 200 (20.0%)
ce_t050_n128	200	53 / 200 (26.5%)	39 / 200 (19.5%)
ce_t050_n512	200	54 / 200 (27.0%)	42 / 200 (21.0%)

Recovered means released decoding produced a valid selected molecule where strict decoding did not. Largest-component selection is the saved released-selection flag. Both fractions divide by all requests, not only valid molecules, and can overlap. CSV/JSON preserve the per-seed counts; these descriptive counts do not isolate a causal effect of repair.

Lexical flags are limited diagnostics, not a complete SAFE/SMILES parser or a causal explanation of quality. Each count uses all 100 requested rows, including decode failures and repeats; no deduplication across seeds or configurations. Fewer flags need not improve QED, SA, validity or quality. The finite-state oracle motivated resolution testing but does not prove gains for learned, temperature-0.5 molecular models.

The original complete V10 report remains unchanged and contains all configurations, generation identities, device mappings and per-run metrics. This appendix hashes that report, its PDF/CSV, raw rows, summaries, controllers, prospective/prior evidence and the exact reused lexical/helper source. JSON/CSV retain exact values and counts. No new molecular sampling, chemistry rescoring, checkpoint load or automatic promotion occurred here.